

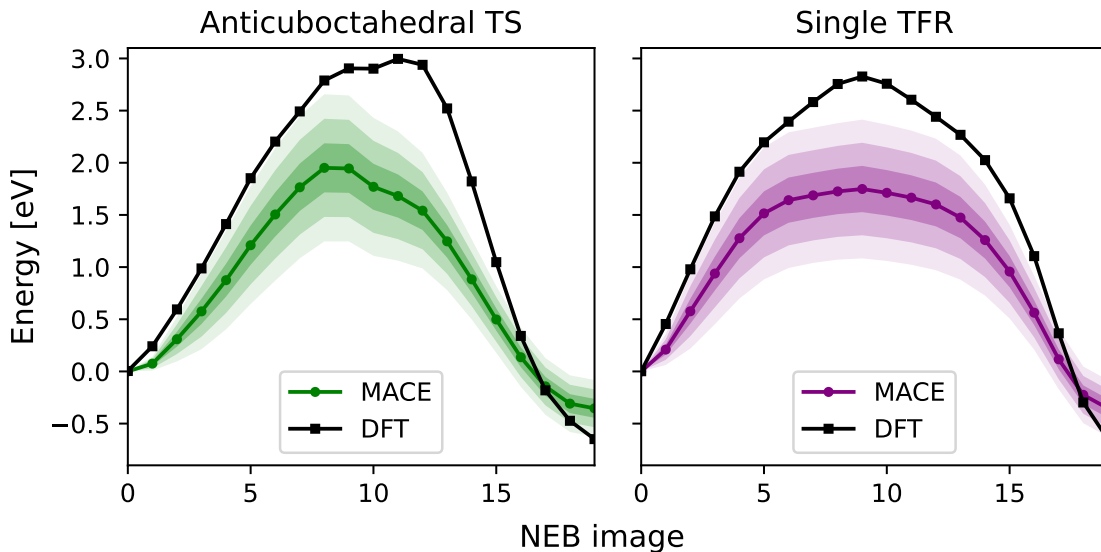


Figure S76: Relative energies predicted by MACE-MP-0b3 along the NEB barriers obtained for the anticuboctahedral transition state and single triangular face rotation pathways suggested for carborane rearrangement. Estimated error bounds are shaded along the plot, where 1σ , 2σ , and 3σ bounds are shown in successively lighter shades. DFT reference data (that is not entirely consistent with the MPtrj dataset) is shown in black.

D.4.2 Calibrating uncertainty for CO oxidation on Cu

To highlight the potential of classification on calibrating force uncertainty in specific cases, we evaluate the uncertainty of forces and energy differences along the NEB paths for CO oxidation on the (111) and (211) surfaces, where LLPR provides only qualitative uncertainty estimates. For calibration, we select 10 uniformly spaced images along each NEB path rather than using random sampling.

Figure S78 compares the predicted force errors against DFT references and illustrates the NEB barrier paths for two different surfaces, with error bars derived from LLPR uncertainty, original CP, and class-based CP.

After calibration, the predicted force errors align more closely with the ground truth for both surfaces. However, when applying original CP to all sites on the (111) surface, force uncertainties remain inaccurate due to the predominance of “bulk” copper surface sites in the calibration set, as indicated by the brown

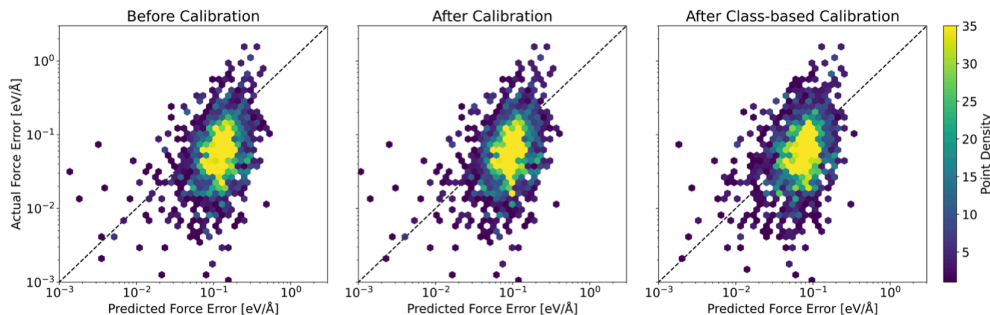


Figure S77: Force errors on the validation set for Large MACE-MP0B3. Quantile in simple CP is $\hat{q} = 0.76$. Quantile for class-based CP with 6 classes ranges from 0.5 to 2.0 and are of similar range with increasing number of clusters.

atoms in the right panels of Figure S78. This imbalance leads to overestimated quantiles and inflated errors for high-error sites. By incorporating classification, the predicted errors for outlier sites better match the true errors, as shown in the left panel of the figure.

For uncertainties in energy differences along the NEB paths (right panel of Figure S78), original CP provides accurate calibration based on LLPR uncertainties. Class-based CP offers a slightly more refined calibration, particularly for the (111) surface.

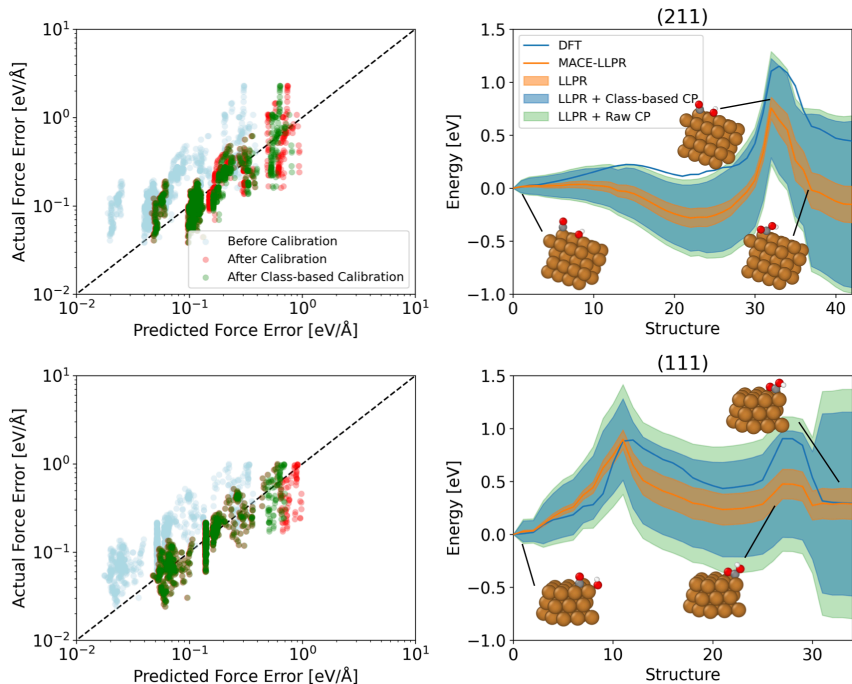


Figure S78: MACE-predicted forces and relative energies along the NEB barrier paths for CO oxidation on the Cu(211) (top) and Cu(111) (bottom) surfaces. The estimated force and NEB barrier path errors from MACE-LLPR uncertainty, after calibration and class-based calibration, are shown in different colors. The DFT reference data for the NEB barrier path is shown in blue.

D.4.3 Calibrating uncertainty for carborane rearrangement

Next, we further evaluate the robustness of the proposed method on a more challenging system: carborane rearrangement. We selected Anticuboctahedral TS and single TFR cases as their results are representative. Calibration samples are selected using the same scheme as in the previous example. We set $\alpha = 0.2$, as this choice enables meaningful variation in quantiles across distinct classes during class-based CP calibration.

As shown in the left panels of Figure S79, the LLPR method slightly underestimates the actual force error in both cases, while calibration improves alignment with the true error. Notably, classification plays a crucial role in achieving robust calibration, as it accounts for the coexistence of both accurate and underestimated LLPR uncertainties within the system. This effect is particularly evident in the isomerization reaction of TFR (bottom), where separate calibration for different subsets of observations is necessary. This underscores the potential application of class-based calibration in active learning for chemically diverse systems, where both severe under- and overestimation of force uncertainties occur during iterative fine-tuning.

However, as shown in the right panels of Figure S79, while calibration significantly improves force error estimates and classification enhances robustness, this effect does not directly carry over to the calibration of energy differences along the NEB path. This discrepancy arises from the strong symmetry in system movement, which leads to error cancellation when summing uncertainty contributions from individual atoms. We speculate that this issue could be mitigated by performing calibration directly on energy rather than force, particularly for energy-related properties.

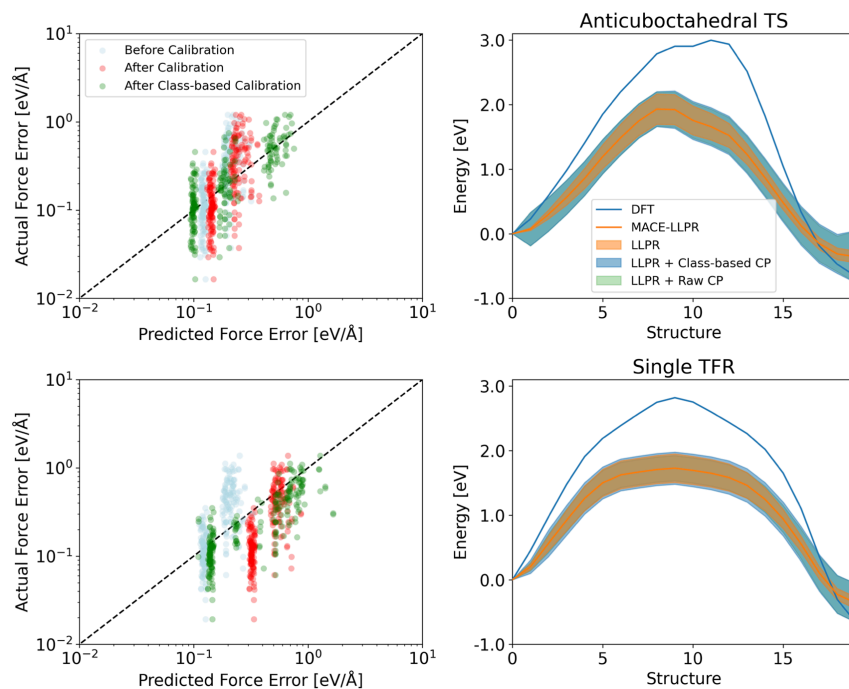


Figure S79: MACE-predicted forces and relative energies along the NEB barrier paths for carborane rearrangement in Anticuboctahedral TS and single TFR. The estimated force and NEB barrier path errors from MACE-LLPR uncertainty, after calibration and class-based calibration, are shown in different colors. The DFT reference data for the NEB barrier path is shown in blue.